

**Dr. Babasaheb Ambedkar Marathwada University
Chhatrapati Sambhajinagar- 431001**



Four Year UG Engineering/Technology Programme

**Syllabus Structure of SY Electrical and Electronics Engineering/
Electrical Engineering / Electrical, Electronics and Power (EEE/EE/EEP)**

(AS PER NEP-2020)

Effective from 2025-26

FACULTY OF ENGINEERING AND TECHNOLOGY Syllabus Structure of SY Electrical Engineering / Electrical and Electronics Engineering/ Electrical, Electronics and Power (EE/EEE/EEP) w.e.f.2025-26 (As Per NEP-2020)										
Semester–III										
Sr. No.	Course Category	Course Code	Course Title	Contact Hours per week		Credits		Scheme of Examination		
				Theory	Practical	Theory	Practical	CIA	SEE	Total
1	Program Core Course (PCC)	PCCT-1	Transformer and DC Machines	3	----	3	----	40	60	100
		PCCP-1	Transformer and DC Machines Lab	----	2	----	1	20	30	50
		PCCT-2	Electrical Measuring Techniques	2	----	2	----	20	30	50
		PCCP-2	Electrical Measuring Techniques Lab	---	2	--	1	20	30	50
		PCCT-3	Electronics Devices & Circuits	2	----	2	----	20	30	50
		PCCP-3	Electronics Devices & Circuits Lab	---	2	--	1	20	30	50
2	Multidisciplinary Minor (MDM)	MDMT-1	As per Basket	2	----	2	---	20	30	50
3	Open Elective (OE) to be chosen compulsorily from the faculty other than Major Discipline	OET-1	As per Basket	2	----	2	----	20	30	50
		OET-2	As per Basket	2	----	2	----	20	30	50
4	Entrepreneurship/ Economics/ Management Courses		Theory-7 One Theory course form the following Basket 1. Engineering Economics 2. Entrepreneurship Development 3. Industrial Management	2	----	2	----	20	30	50
5	Value Education Course (VEC)	VECT-1	Theory-8 (Universal Human Values/ Environmental Studies)	2	---	2	--	20	30	50
6	Comm. Engg. Project/Field Project(FP)	FP-1	Field Project	----	4	----	2	20	30	50
				17	10	17	5	260	390	650

FACULTY OF ENGINEERING AND TECHNOLOGY Syllabus Structure of SY Electrical Engineering / Electrical and Electronics Engineering/ Electrical, Electronics and Power (EE/EEE/EEP) w.e.f.2025-26 (As Per NEP-2020)										
Semester-IV										
Sr. No.	Course Category	Course Code	Course Title	Contact Hours per week		Credits		Scheme of Examination		
				Theory	Practical	Theory	Practical	CIA	SEE	Total
1	Program Core Course(PCC)	PCCT-1	AC Machines	3		3		40	60	100
		PCCP-1	AC Machines Lab		2		1	20	30	50
		PCCT-2	Electrical Power Transmission and Distribution	2		2		20	30	50
		PCCP-2	Electrical Power Transmission and Distribution Lab		2		1	20	30	50
		PCCT-3	Network Analysis	2		2		20	30	50
		PCCP-3	Network Analysis Lab		2		1	20	30	50
2	(MDM) Multidisciplinary Minor	MDMT-2	As per Basket	2		2		20	30	50
3	(OE) to be chosen compulsorily from the faculty other than Major Discipline Open Elective	OET-3	As per Basket	2		2		20	30	50
4	Vocational and Skill Enhancement Course (VSEC)	VSECT-2	Programme specific Skills (PLC Programming)		4		2	20	30	50
5	Ability Enhancement Course (AEC)	AEC-2	Theory-6 Modern Indian Languages (Marathi, Hindi, Sanskrit, Urdu, Pali, Arabic)	2		2		20	30	50
6	Economics/ Management Courses/ Entrepreneurship		Theory-7: One Theory course form the following Basket Engineering Economics Entrepreneurship Development Industrial Management	2		2		20	30	50
7	Value Education Course (VEC)	VECT-2	Theory-8 (Universal Human Values/ Environmental Studies)	2		2		20	30	50
				17	10	17	5	260	390	650
1	Honors	ETH 3 to 8	As per proposed list of Honor Courses	3		3		40	60	100
2	Double Minor	ETDM 3 to 8	As per proposed list of Double Minor Courses	3		3		40	60	100

Baskets of Program Elective Course

Semester / Domains	Energy & Power control	Electronics	Transmission & Distribution
V-PEC-1	Conventional Energy sources	Digital Electronics	Electrical Safety
V-PEC-2	Renewable Energy Sources	Analog Integrated Circuit	Industrial Automation
VI-PEC-3	Power Electronics	Embedded Systems	Electrical Traction
VII-PEC-4	Power Quality	Illumination Engineering	Energy conservation and Audit

Basket of Open Electives Courses

Semesters / Domains	FinTech	Multimedia Forensic	Social Sciences
III-OE1	Introduction to Financial Market	Introduction to Digital Forensics	Indian Constitution
III-OE2	Risk Management and Compliances	Cyber Security and cyber law	Emotional Intelligence
IV- OE3	Data Analytics for Finance	Incident Response Management	Introduction to Sociology
V-OE4	Business Intelligence	Practical Case Studies and Investigations	Indian Economy

Basket of Multi-disciplinary Minor

Semester	Mechanical	E&TC / ETX/ EC	EE / EEE/ EEP	Civil	CSE/AIIML/ AIDS
Domain	Industrial Automation	IOT	EV	Smart Cities	AIIML
III MDMT-1	Robotics& Automation	IoT Fundamentals	Introduction to Electric Vehicles (EVs)		
IV MDMT-2	Robotics Kinematics and Dynamics	IoT Hardware & Sensors	EV Technology and Components		
V MDMT-3	Sensors and Actuators for Robotics	IoT Connectivity & Communication	EV Infrastructure and Charging		
VI MDMT-4	Artificial Intelligence and Machine Learning in Robotics	IoT Data Processing & Analytics	EV Market and Industry Trends		
VII MDMT-5	PLC and SCADA	Hands on Applications & Project Developments	Hands on EV Design		

Exit Course

Exit option: Award of UG Diploma in Major with 88 credits and an additional 8 credits from following Exit Courses				
Sr. No.	Course Code	Course Title	Mode	Credits
1		Maintenance and Servicing of Electrical & Electronics Equipment	Online/offline certification Course	8
OR				
2		Certificate Programme (C, C++ language learned in Sem-1 and/or Sem-2)		8

--	--	--	--	--

**SY BE /B.Tech. in Electrical and Electronics Engineering (EEE) / Electrical Engineering (EE) /
Electrical, Electronics and Power Engineering (EEP) with Double Minor Degree
W.e.f.2025-26 (As Per NEP-2020)**

1. It is required to complete SIX courses (each of 3 credits) from ONLINE/OFFLINE platform to earn total of 18 credits under Honor/ Double Minor (DM) certification.
2. It is optional for students to take Honors/Double Minor (Multidisciplinary and Specialization minors).
3. Each Eligible Student can opt for maximum one honor or Minor.
4. Student must complete and earn the credits for all the six courses starting from Second Year First semester (3rd semester) to Final Year Second Semester (8th semester).
5. Basket of the DM courses and respective semester is mentioned in the following table.

Basket of double Minor Courses

Sr. No.	Semester	Course	Code
1	III	DM-I	ETDM-1
2	IV	DM-II	ETDM-2
3	V	DM-III	ETDM-3
4	VI	DM-IV	ETDM-4
5	VII	DM-V	ETDM-5
6	VII	DM-VII	ETDM-6

**SY BE /B.Tech. in Electrical and Electronics Engineering (EEE) / Electrical Engineering (EE) /
Electrical, Electronics and Power Engineering (EEP) with Honor Degree
w.e.f.2025-26 (As Per NEP-2020)**

1. It is required to complete SIX courses (each of 3 credits) from ONLINE/OFFLINE platform to earn total of 18 credits under Honor/ Double Minor (DM) certification.
2. It is optional for students to take Honors/Double Minor (Multidisciplinary and Specialization minors).
3. Each Eligible Student can opt for maximum one honor or Minor.
4. Student must complete and earn the credits for all the six courses starting from Second Year First semester (3rd semester) to Final Year Second Semester (8th semester).
5. 3. Basket of the Honor courses and respective semester is mentioned in the following table.

Basket of Honor Courses

Sr. No.	Semester	Course	Code
1	III	Honor-I	ETH-1
2	IV	Honor -II	ETH-2
3	V	Honor -III	ETH-3
4	VI	Honor -IV	ETH-4
5	VII	Honor -V	ETH-5
6	VII	Honor -VII	ETH-6

CourseCode:PCCT-1 Course:Transformer and DC Machines Credits: 3 Theory :3Hrs./ Week ; 45 Hrs / Semester					
Prerequisites	Fundamentals of Electrical & Electronics Engineering				
Objectives	1. To introduce the basic working principles of single phase transformers and three phase transformers with their applications. 2. To make the students to know about the functionality of DC Machines with their classification and applications.				
Course Outcomes	Student understood the transformers working under various load conditions, their performance parameters and autotransformers. Student got the idea of constructional details, working and modes of operation of single phase transformers and different connections and three winding transformers. Student understood the constructional details about DC Machines. Student got the idea of working of DC Generator and DC Motor. Student understood the different types of starters, speed control, tests & applications. Student understood the constructional details and operation of Special Purpose DC Machines.				
Unit-I	A: Transformer basics: Transformer working & phasor diagrams- on-load and on no-load, Equivalent circuits, Equivalent circuit parameters calculation from the test data, Losses, Efficiency, Ratings. Open circuit and short circuit tests, Polarity test, Parallel operation, and conditions to be satisfied, Load sharing under various conditions. Autotransformers, Comparison of autotransformers with two winding transformer with respect to saving of copper and size, Ratings, Applications. (9Hrs.)				
	B: Three phase transformers & their connections: Construction, types & comparison with a bank of three single phase transformers, Standard connections, phasor groups as per clock notations, polarity test, Efficiency and regulations by direct and indirect method, Parallel operation of transformer Scott connections and 'V'-connections, tertiary windings, three winding transformers. (9Hrs.)				
Unit-II	A: Construction of D.C. Machine: Construction, main parts, magnetic circuit, typical flux path, commutator and brush assembly, Armature winding: Simple lap and wave winding, (6Hrs.)				
	B: DC Generator and DC motor: Generator and motor action, e.m.f equation, types, characteristics, applications, torque equation of motor, significance of back e.m.f. working at no-load and on-load. Losses, Power stages, Efficiency. Armature reaction (Descriptive treatment). Commutation, Causes of bad commutation and remedies (9Hrs.)				
Unit-III	A: Starting Control and Testing of DC Motor: Necessity of starters, types of starters for series and shunt motor, solid state starters, speed control methods, tests, applications (6 Hrs.)				
	B: Special Purpose DC Machines: Construction details and principle of operation of Brush less DC motor, stepper motor, PMDC motor, DC servo motor. (6 Hrs.)				
References	Sr.No.	Title	Author	Publication	Edition
	1	Electrical Technology, Vol II	B. L. Theraja	S.Chand Publishing	1 st Multicolour Edition
	2	Electrical Technology	Edward Hughes	Pearson Education.	-
	3	Electrical Machine	S. K. Bhattacharya	Tata Mc Graw Hill	2 nd Edition

Faculty of Science & Technology Syllabus of SY Electrical Engineering / Electrical and Electronics Engineering/ Electrical, Electronics and Power (EE/EEE/EEP) w.e.f.2025-26 (As Per NEP-2020)(Semester III)					
CourseCode: PCCP-1 Course: Transformer and DC Machines Lab Credits: 1 Practical:02Hrs./Week; 30Hrs/Semester					
Prerequisites		Basic Electrical Engineering			
Objectives	1. 2.	To provide practical experience with the construction, operation, and testing of transformers & dc machines, understanding of their principles. To learn different types of windings, characteristics, losses, efficiency, and testing methods.			
Course Outcomes	1. 2. 3. 4.	Students will be able to: Understand various tests on Transformer. Apply exact experimental setup for performance evaluation of machines. Understand various tests on DC machines. Interpret obtained results to reach appropriate conclusion.			
List of Experiments	1. 2. 3. 4. 5. 6. 7. 8. 9. 10. 11. 12. 13. 14. 15.	To find equivalent circuit parameters from O.C and S.C Test on single phase transformer. To study the parallel operation of single phase transformers. To study Scott connection. To study Load test on single phase transformer. To study Swmpner's test on two single-phase transformers. To separate core losses of single-phase transformer at no-load. To obtain Efficiency and regulation of three phase transformer by direct loading. To perform Polarity test and ratio test on three phase transformer. To perform Load test on DC shunt motor. To perform Load test on DC series motor. To obtain the Magnetizing Characteristics of separately excited D.C Generator. To study Speed Control of DC Shunt Motor (Armature and Field Control). To obtain the efficiency of D.C machine using Swinburne's test. To study Hopkinson's Test. To study the working of special purpose motors.			
	4	Electrical Machines	Nagrath & Kothari.	Tata Mc Graw Hill	-
	5	Electrical Machines	Bhag S Guru, Husein R. Hizirolu	Oxford University Press	-
	6	Electrical Machines- I and II	K Krishna Reddy	SCITECH Publications (India) Pvt. Ltd. Chennai	-

FACULTY OF ENGINEERING AND TECHNOLOGY Syllabus Structure of SY Electrical Engineering / Electrical and Electronics Engineering/ Electrical, Electronics and Power (EE/EEE/EEP) SEM-III	
Course Code: PCCT-3 Course: Electronics Devices & Circuits Credits: 2 Theory: 2Hrs./ Week;30 Hrs / Semester	
Prerequisites	Knowledge of Basic Electronics Engineering
Objectives	1. To provide students with a foundational understanding of basic electronics, including the principles of voltage, current, resistance, capacitance, and inductance in different electronic components. 2. Teach students the principles of semiconductor physics, focusing on the working of diodes, transistors (BJTs, FETs). 3. To give students an understanding of amplifiers (both small and large signal), their operation, frequency response, and design principles. 4. To introduce students to the concept of feedback in amplifiers and its effect on performance. 5. To explain the theory, operation, and applications of operational amplifiers (Op-Amps) in different configurations, such as inverting, non-inverting, integrator, differentiator, and filters.
Course Outcomes	1. Demonstrate knowledge of the different electronic components (resistors, capacitors, inductors, diodes, transistors, etc.). 2. Understand the working principles of semiconductor materials and devices, including p-n junctions, diodes, transistors (BJT, FET), and their applications in electronic circuits. 3. Analyze and design electronic circuits using diodes, transistors, and operational amplifiers for both analog and digital applications. 4. Understand and design amplifiers (common emitter, common base, common collector, op-amp circuits) and oscillators.
Unit-I	A : Rectifiers and Filters Half wave rectifier, full wave rectifier, bridge rectifier, PIV, efficiency, ripple factor. TUF analysis, Ripple Factor calculation for C, L, LC, voltage multiplier, with diodes, regulation with and without filter, load and line regulation. (5 Hrs)
	B : Theory of Junction Transistors and field Effect Transistor Transistor action – Transistor parameter, Transistor current components, Transistor as an amplifier and as switch , collector efficiency, early effect in transistor, Junction FET operation, Enhancement MOSFET, depletion MOSFET, comparison. JFET and MOSFET. (5 Hrs)
	A : Bias Stability and Device Stabilization Transistor Biasing : Location of Q point, fixed bias circuit, collector to base circuit, self-bias circuit, graphical DC bias analysis, design of all biasing circuits, FET biasing, : self-biasing, voltage feedback biasing. (6 Hrs)

Unit-II	B : Amplifiers Frequency Response, RC coupled and transformer coupled amplifier, single stage and multistage amplifier, wide band amplifier, cascade amplifiers, feedback amplifiers, positive and negative feedback, current and voltage feedback. (6 Hrs)
---------	---

Unit-III	A : Analysis of CE,CC,CB circuits, voltage gain, current gain, input/output impedance, dependence on source and load impedance, emitter follower analysis, boot strapping in emitter follower. (5 Hrs)
	B : Power Amplifiers Types: Class A, Class B, Class AB, and Class C, capacitor coupled quasi complimentary, push pull, expression for efficiency of Class A class B amplifier, distortion in amplifier (5 Hrs)

References	Sr. No.	Title	Author	Edition
	1	A Course in Electrical Engineering Materials	S. P. Seth, Dhanpat Rai and Sons	3 th Edition
	2	Electrical Engineering Materials	K. B. Raina & S. K. Bhattacharya, S. K. Kataria & Sons,	4 th Edition
	3	Introduction to Nanotechnology	Charles P. Poole, Jr. Frank & J. Ownes	1 th Edition
	4	Introduction to Nano Science & Technology	Chattopadhyaya	2 th Edition
	5	"Electronic Devices"	Thomas L. Floyd	10th Edition (2015)

FACULTY OF ENGINEERING AND TECHNOLOGY Syllabus Structure of SY Electrical Engineering / Electrical and Electronics Engineering/ Electrical, Electronics and Power (EE/EEE/EEP) SEM-IV	
Course Code: PCCP-3 Course: Electronics Devices & Circuits Lab Credits: 1 Theory: 2 Hrs./ Week; 30 Hrs / Semester	
Prerequisites	Machines Fundamental

Objectives	8. To connect theoretical knowledge with practical applications, such as in communication systems, power electronics, signal processing, and instrumentation. 7.To develop students' ability to apply problem-solving techniques and use tools (such as circuit simulation software and laboratory equipment) to analyze and design electronic circuits effectively.
Course Outcomes	5. Understand the basic principles and circuits used in communication systems such as modulators, demodulators, and RF amplifiers. 6. Understand the ethical considerations in designing electronic circuits and the environmental impact of electronic waste, ensuring sustainable design practices..
Experiments	Any 12 practical to be conducted on 1. Determination of ripple factor. PIV. Efficiency regulation factor of half wave and full wave amplifiers (with and without capacitors). 2. Comparative study of fixed and self-biased circuits. 3. Study of transformer coupled power amplifiers. 4. Study of transistor characteristics in CE configuration and Determination of h-parameters graphically. 5. Comparison of frequency response of RC coupled amplifiers with feedback and without feedback (comparison of Gain and Bandwidth). 6. Determination of Voltage gain, current gain, input and output impedance of FET amplifiers. 7. Designing and testing of RC phase shift oscillator. 8. Study and frequency calculation of Hartley oscillator. 9. Study and frequency calculation of Colpitts oscillator. 10. Study of class a, class B, class AB amplifiers. 11. To Study Zener Diode Characteristics and Voltage Regulation 12. Study the input and output characteristics of a Bipolar Junction Transistor (BJT) in different configurations. 13. Study the frequency response and gain of a common emitter amplifier. 14. Design and test a power amplifier circuit. 15. Study the time constant of RC and RL circuits.

FACULTY OF ENGINEERING AND TECHNOLOGY Syllabus Structure of SY Electrical Engineering / Electrical and Electronics Engineering/ Electrical, Electronics and Power (EE/EEE/EEP)	
Course Code: MDMT-1 Course: Introduction to Electric Vehicles (EVs) Credits: 2 Theory :2Hrs./ Week ; 30 Hrs / Semester	
Prerequisites	Basics of Electrical and Electronics, Engineering Chemistry, Engineering Systems.
Objectives	To understand the basic technologies used in e-vehicles and the necessary advancements in it.
Course Outcomes	1. Understand the basics related to e-vehicle. 2. Classify the different hybrid vehicles. 3. Evaluate the different driving systems for e-vehicles. 4. Analyze EV benefits and challenges.
Unit-I	Introduction to Hybrid Electric Vehicles: History of hybrid and electric vehicles, social and environmental importance of hybrid and electric vehicles, impact of modern drive-trains on energy supplies. Conventional Vehicles: Basics of vehicle performance, vehicle power source characterization, transmission characteristics, mathematical models to describe vehicle performance.
Unit-II	Hybrid Electric Drive-trains: Basic concept of hybrid traction, introduction to various hybrid drive-train topologies, power flow control in hybrid drive-train topologies, fuel efficiency analysis.
Unit-III	Electric Drive-trains: Basic concept of electric traction, introduction to various electric drive-train topologies, power flow control in electric drive-train topologies, fuel efficiency analysis.
References	1. Iqbal Hussein, Electric and Hybrid Vehicles: Design Fundamentals, CRC Press, 2003. 2. Mehrdad Ehsani, Yimi Gao, Sebastian E. Gay, Ali Emadi, Modern Electric, Hybrid Electric and Fuel Cell Vehicles: Fundamentals, Theory and Design, CRC Press, 2004. 3. James Larminie, John Lowry, Electric Vehicle Technology Explained, Wiley, 2003 4. Modern Electric, Hybrid Electric, and Fuel Cell Vehicles Third Edition, Mehrdad Ehsani Yimin Gao Stefano Longo Kambiz M. Ebrahimi

FACULTY OF ENGINEERING AND TECHNOLOGY
Syllabus Structure of SY Electrical Engineering / Electrical and Electronics Engineering/ Electrical, Electronics and Power (EE/EEE/EEP)
w.e.f.2025-26

Course Code: OET-1

Course: Introduction to Financial Market

Credits: 2

Theory: 2 Hrs./ Week; 30 Hrs / Semester

Prerequisites	Basic Knowledge of Market.
Objectives	1. Introduce students to the structure and functioning of financial markets. 2. Provide an understanding of financial marketing as a specialized area of marketing. 3. Study the factors influencing consumer behavior in financial services, including psychological, social, and economic factors. 4. Introduce the impact of digital transformation on financial services marketing. 5. Examine strategies that financial institutions (banks, insurance companies, asset management firms) use to market their products and services.
Course Outcomes	1. Understanding the Fundamentals of Financial Markets and Instruments. 2. Apply Marketing Principles to Financial Products and Services. 3. Analyze Consumer Behavior in Financial Services. 4. Develop a Digital Marketing Strategy for Financial Services.
Unit-I	Business Management : Concept of Management-Definitions, Characteristics, Management & Administration, Different Concepts, Management by Communication, Management by Systems, Management by Results, Management by Participation, Management by Motivation, MBE, MBO –Functions of management. Planning : Meaning, Nature, Definition, significance, Process, Types, Importance, Principles, Steps – Planning and forecasting, Planning and Decision, Making and Planning and Control (10 hr)
Unit-II	Organizing :definitions, Principles of Organization, Formal & informal Organizations, Steps, Importance, Organization Development, Problems in Organization, Delegation - Centralization and Decentralization–Span of Control, Types or Forms of Organization - Organizational goals – Determination of Organizational goals and Determinants of Organizational goals (10 hr)
Unit-III	Directing :Meaning , Definitions, Nature of Directions, Elements of Direction, Directing and Other Managerial Activities, Importance of Direction-Orders and Instructions, Supervision, Essentials of Effective Supervision and Principles of Direction. Leading: Meaning, Functions, Qualities of good leader – Leadership Styles - Motivation-Meaning, Nature Definitions, Benefits, and Modern Theories of Motivation

	– McGregor’s Theory, Maslow’s hierarchy of Needs Theory - Herzberg’s Theory. (10 hr)				
References	Sr. No.	Title	Author	Publication	Edition
	1	Marketing of Financial Services	V. S. Ramaswamy & S. Namakumari	Macmillan India	1 st Edition
	2	Principles & Practice of Management	Prasad L.M.,	Sultan Chand & Sons	-
	3	Introduction to Financial Markets: A Global Perspective	M. Jayant R. Ghosh	Springer	1 th Edition
	4	Financial Markets and Institutions	Frederic S. Mishkin & Stanley G. Eakins	Pearson	9 th Edition
	5	Principles of Financial Marketing	Martin P. S. S. K. Agrawal	McGraw-Hill	1 st Edition

FACULTY OF ENGINEERING AND TECHNOLOGY Syllabus Structure of SY Electrical Engineering / Electrical and Electronics Engineering/ Electrical, Electronics and Power (EE/EEE/EEP) w.e.f.2025-26	
Course Code: OET-1 Course: Introduction to Digital Forensics Credits: 2 Theory: 2 Hrs./ Week; 30 Hrs / Semester	
Prerequisites	Knowledge of Basics of Computer, General Cyber Information.
Objectives	1. To provide students with a comprehensive understanding of the basic concepts of digital forensics, including its scope, objectives, and importance in modern cyber security and legal contexts. 2. To educate students about the legal framework governing digital forensics and the ethical considerations associated with handling digital evidence. 3. To familiarize students with the various tools and techniques used in digital forensics to acquire, examine, and analyze digital evidence from computers, mobile devices, networks, and cloud systems. 4. To provide students with the skills to properly acquire and preserve digital evidence in a manner that maintains its integrity and admissibility in a court of law.

	5. To equip students with the knowledge required to investigate and analyze digital evidence from computers and mobile devices, which are commonly involved in cybercrimes.
Course Outcomes	1. Understand the legal aspects of forensics 2. Analyze Legal and Ethical Considerations in Digital Forensics 3. Utilize Digital Forensic Tools and Techniques 4. Investigate Network-Based Incidents
Unit-I	Introduction to Digital Forensics: Definition, scope, and importance of digital forensics History and evolution of digital forensics Legal and ethical issues in digital forensics The role of digital forensics in cybercrime investigation Types of digital forensics: Computer forensics, network forensics, mobile forensics, cloud forensics, etc. Digital evidence: Types and classification, Chain of custody and handling digital evidence, Forensic analysis process: Acquisition, examination, analysis, and reporting
Unit-II	Legal and Ethical Aspects of Digital Forensics : Legal Framework for Digital Forensics, Overview of cybercrime laws (e.g., Information Technology Act, 2000 in India, or similar laws in other countries), International legal considerations in digital forensics (e.g., data privacy, cross-border data issues), Legal requirements for digital evidence handling and admissibility in court, Ethical Issues in Digital Forensics : Ethical considerations in the forensic examination of digital evidence, Privacy concerns and ethical dilemmas in digital investigations, Professional ethics and code of conduct for digital forensic professionals

Unit-III	Digital Forensics Tools and Techniques : Forensic Tools Overview, Overview of common forensic tools and software (e.g., EnCase, FTK, Autopsy, Sleuth Kit, etc.), Free and open-source tools for digital forensics, Selection of tools based on case needs and evidence types, Techniques for Data Acquisition : Methods of acquiring digital evidence: Live data acquisition vs. dead imaging, Tools and techniques for creating bit-by-bit copies of digital storage media, Handling encrypted data and password-protected devices. Forensic Imaging and Duplication: Creating forensic images (write-blocking techniques, hashing, etc.), Ensuring integrity of data during duplication, Verifying the authenticity of evidence (hashing, checksums, etc.)				
	Sr. No.	Title	Author	Publication	Edition

References	1	"Digital Evidence and Computer Crime: Forensic Science, Computers, and the Internet"	Eoghan Casey	Academic Press	3 th Edition
	2	Guide to Computer Forensics and Investigations"	Bill Nelson, Amelia Phillips, Christopher Steuart	Cengage Learning	5 th Edition
	3	"Computer Forensics: Investigating Network Intrusions and Cybercrime"	EC-Council	EC-Council Press	2 th Edition
	4	"Digital Forensics and Incident Response"	Jason Luttgens, Matthew Pepe, Kevin Mandia	McGraw-Hill Education	1st Edition

FACULTY OF ENGINEERING AND TECHNOLOGY Syllabus Structure of SY Electrical Engineering / Electrical and Electronics Engineering/ Electrical, Electronics and Power (EE/EEE/EEP) w.e.f.2025-26	
Course Code: OET-1 Course: Indian Constitution Credits: 2 Theory: 2 Hrs./ Week; 30 Hrs / Semester	
Prerequisites	Basic Knowledge of Indian history.
Objectives	1. To provide students with a historical perspective on the making of the Indian Constitution, including the pre-independence constitutional developments and the influence of various British and Indian legal documents. 2. It will explore various aspects of the Indian political and legal system from a historical perspective highlighting the various events that led to the making of the Indian Constitution. 3. To comprehend the Preamble of the Indian Constitution, which outlines the objectives of the Constitution, such as justice, liberty, equality, and fraternity? 4. To understand the structure of the Indian government, including the Union Government (Executive, Legislature, Judiciary) and the State Governments, and the distribution of powers between the center and states (federal structure). 5. To familiarize students with the Fundamental Rights (Part III of the Constitution), which guarantee basic human rights to citizens, and understand their scope, limitations, and legal protection.

Course Outcomes	6. List salient features and characteristics of the constitution of India. 7. Follow fundamental rights and duties as responsible citizen of the country. 8. Analyze major constitutional amendments in the constitution. 9. Follow procedure to cast vote using voter-id.
Unit-I	Constitution and Preamble Meaning of the constitution of India. Historical perspectives of the Constitution of India, Salient features and characteristics of the Constitution of India, Preamble of the Constitution of India. Fundamental Rights under Part-III. Fundamental duties and their significance under part-IV-A. Relevance of Directive Principles of State Policy under part-IV A. '(12 Hrs)
Unit-II	Governance and Amendments : Amendment procedure of the Constitution and their types - simple and special procedures, The Principle of Federalism and its contemporary significance along with special committees that were setup, Major Constitutional Amendment procedure - 1st, 7th, 42nd, 44th, 73rd & 74th, 76th, 86th, 52nd & 91st, 102 nd (9 Hrs)

Unit-III	Electoral Literacy and Voter's Education : Electoral rights , Electoral process of registration, Ethical electoral participation, Motivation and facilitation for electoral participation, Voter's guide, Prospective empowered voter, Voting procedure, Voter awareness, Voter online registration https:// www.ceodelhi.gov.in/ELCdetails.aspx '(9 Hrs)				
References	Sr. No.	Title	Author	Publication	Edition
	1	P.M.Bakshi	The Constitution of India	Universal Law Publishing, New Delhi 15th edition, 2018, ISBN: 9386515105 (Check the new edition)	15 th Edition
	2	D.D.Basu	Introduction to Indian Constitution	Lexis Nexis Publisher, New Delhi, 2015	-
	3	B. K. Sharma	A. E. Fitzgera Introduction to Constitution of India Id	PHI, New Delhi, 6th edition, 2011	6 th Edition

	4	B.L. Fadia	The Constitution of India	Sahitya Bhawan,Agra, 2017	-
--	---	------------	------------------------------	---------------------------------	---

Faculty of Science & Technology Syllabus of S. Y. B.E./B.TECH.(Computer Science and Engineering; AI&ML; Electrical Engineering, Artificial Intelligence, Information Technology, Electrical and Electronics Engineering, Electronics and Telecommunication Engineering) (Semester III)	
Course Code: OET-2 Course: Risk Management and Compliances Credits: 2 Theory :2Hrs./ Week ; 30 Hrs / Semester	
Prerequisites	Basic Sciences
Objectives	0. To provide knowledge of Risk Management. 1. To understand Potential Risks. 2. To study Assess and Prioritize Risks. 3. To Develop Risk Response Strategies. 4. Monitor and Review Risks Continuously & Enhance Decision-Making.
Course Outcomes	Students should Identify and Classify Risks Analyze and Evaluate Risk Scenarios Develop Effective Risk Mitigation Strategies in students. Students able to Apply Risk Management Frameworks and Standards. Communicate Risk Information Effectively.
Unit-I	A) Introduction to Risk Management (05 HRS.) Concept and Nature of Risk, Types and Sources of Risk (Strategic, Operational, Financial, Hazard, Reputational), Risk Management Process and Principles Importance and Benefits of Risk Management, Role of Risk Manager
	B) Risk Assessment and Analysis (05 HRS.) Risk Identification Techniques (Brainstorming, Checklists, Interviews, SWOT),Risk Analysis: Qualitative and Quantitative, Risk Evaluation and Prioritization, Risk Appetite and Tolerance, Tools and Techniques (Risk Matrices, Monte Carlo Simulation, Sensitivity Analysis)
Unit-II	A) Risk Treatment and Mitigation (05 HRS.) Risk Response Strategies (Avoidance, Reduction, Sharing, Acceptance), Designing Risk Mitigation Plans, Implementation of Risk Controls, Cost-Benefit Analysis of Risk Mitigation, Business Continuity and Disaster Recovery Planning.
	B) Compliance Management (05 HRS.) Concept of Compliance and Corporate Governance, Regulatory Environment and Legal Framework (national & international), Key Compliance Requirements (e.g., GDPR, SOX, HIPAA, ISO Standards), Ethics and Integrity in Compliance, Roles of Compliance Officer

	and Internal Audit.				
Unit-III	A) Risk Monitoring, Reporting, And Framework (10 HRS) Risk Monitoring and Review Mechanisms, Risk Reporting and Documentation, Integration of Risk Management with Strategic Planning, Risk Management Frameworks (ISO 31000, COSO ERM, Basel Accords), Case Studies: Risk and Compliance Failures & Successes. B) Write Report on risk Management and present seminar.				
References	Sr. No.	Title	Author	Publication	Edition
	1	Risk Management and Financial Institutions	John C. Hull	Wiley	5th Edition
	2	Principles of Risk Management and Insurance	George E. Rejda, Michael McNamara	Pearson	14th Edition
	3	The Essentials of Risk Management	Michel Crouhy, Dan Galai, Robert Mark	McGraw Hill	2nd Edition

Faculty of Science & Technology Syllabus of S. Y. B.E./B.TECH.(Computer Science and Engineering; AI&ML; Electrical Engineering, Artificial Intelligence, Information Technology, Electrical and Electronics Engineering, Electronics and Telecommunication Engineering) (Semester III)	
Course Code: OET-2 Course: Cyber Security and cyber law Credits: 2 Theory :2Hrs./ Week ; 30 Hrs / Semester	
Prerequisites	Basic Sciences
Objectives	5. To Understand the Fundamentals of Cyber security. 6. Learn to Identify and Prevent Cyber Threats 7. To Gain Knowledge of Cyber Laws and Regulations. 8. To Promote Ethical and Legal Use of Technology. 9. Prepare for Careers in Cyber security and Legal Compliance.
Course Outcomes	Understand Core Concepts of Cyber security. Analyze Cyber Threats and Apply Security Measures. Interpret and Apply Cyber Laws. Demonstrate Ethical and Legal Digital Behavior. Develop Basic Skills for Cyber security Roles.
Unit-I	A) Introduction to Cyber security (05 HRS.) Overview of Cyber security Importance and Scope in the Digital World Types of Cyber Threats (Malware, Phishing, Hacking, etc.) Vulnerabilities and Attack Vectors Cyber security in Business, Government, and Personal Use
	B) Security Technologies and Measures (05 HRS.) Network Security Fundamentals (Firewalls, IDS/IPS) Cryptography: Symmetric and Asymmetric Encryption Authentication and Access Control Mechanisms Cloud Security and Mobile Device Security Data Backup, Disaster Recovery, and Business Continuity
Unit-II	A) Cybercrime and Investigation (05 HRS.) Types of Cybercrimes: Identity Theft, Cyberbullying, Financial Fraud, Cyber Terrorism and CERTs (Computer Emergency Response Teams) Investigation Procedures Techniques Used by Cybercriminals Role of Cyber Police and Challenges Real-world Cybercrime Case Studies
	B) Introduction to Cyber Laws (05 HRS.) Definition and Scope of Cyber Laws Need for Cyber Law in the Digital Age Legal Recognition of Electronic Records and Signatures Offenses and Penalties under the IT Act Intellectual Property in Cyberspace (Copyright, Trademarks, Patents)

Unit-III	A) Privacy, Ethics, and Global Frameworks (10 HRS) Data Protection and Privacy Laws (GDPR, HIPAA, Indian PDP Bill) Ethical Issues in Cyber security Cyber Ethics, Netiquette, and Responsible Use of Technology International Cyber Laws and Conventions Legal Aspects of Social Media, E-commerce, and Online Banking Future Trends: AI, IoT, and Legal Implications. B) Write Cyber Security & Cyber Laws and present seminar.				
References	Sr. No.	Title	Author	Publication	Edition
	1	Cyber security for Beginners	Raef Meeuwisse	Cyber Simp Ltd.	2nd Edition
	2	Computer Security: Principles and Practice	William Stallings, Lawrie Brown	Pearson Education	4th Edition
	3	Cyber security and Cyber war: What Everyone Needs to Know	P.W. Singer, Allan Friedman	Oxford University Press	1st Edition

Faculty of Science & Technology Syllabus of S. Y. B.E./B.TECH.(Computer Science and Engineering; AI&ML; Electrical Engineering, Artificial Intelligence, Information Technology, Electrical and Electronics Engineering, Electronics and Telecommunication Engineering) (Semester III)	
Course Code: OET-2 Course: Emotional Intelligence Credits: 2 Theory :2Hrs./ Week ; 30 Hrs / Semester	
Prerequisites	Basic Sciences
Objectives	1. To help students recognize their emotions and understand how feelings influence their thoughts, behavior, and academic performance. 2. To enable students to manage stress, control impulses, and respond to challenges with patience and a positive attitude. 3. To teach students to recognize and respect the emotions and perspectives of others, promoting kindness, inclusion, and teamwork. 4. To help students express themselves clearly and respectfully, resolve conflicts peacefully, and build strong peer relationships. 5. To encourage goal-setting, persistence, and a growth mindset, helping students stay motivated and bounce back from setbacks.
Course Outcomes	1. Demonstrate Self-Awareness. 2. Apply Techniques for Emotional Self-Regulation. 3. Exhibit Empathy and Social Sensitivity. 4. Build and Maintain Healthy Relationships. 5. Develop Motivation and a Growth Mindset. 6. Encourage Responsible Decision-Making.
Unit-I	A) Introduction to Emotional Intelligence (05 HRS.) Definition and Historical Background of EI Components of Emotional Intelligence: Self-awareness Self-regulation Motivation Empathy Social skills EI vs IQ: Why Engineers Need Both Importance of EI in Personal and Professional Life EI in Academic Success and Employability
	B) Developing Self-Awareness (05 HRS.) Understanding Emotions and Emotional Triggers Emotional Self-Assessment Techniques Mindfulness and Reflection Practices Building Self-Confidence and Personal Vision Personality Types and Emotional Tendencies (e.g., MBTI, Big Five)
Unit-II	A) Managing Emotions and Stress (05 HRS.) Techniques for Emotional Regulation and Control Managing Anger, Anxiety, and Frustration Time and Stress Management in Academic Life

	Resilience and Adaptability in Engineering Challenges Growth Mindset and Self-Motivation				
	B) Empathy and Social Awareness (05 HRS.) Recognizing Others' Emotions and Non-verbal Cues Empathy in Academic Teams and Project Groups Emotional Dynamics in Multicultural Teams Listening Skills and Perspective-Taking Building Inclusive and Respectful Peer Relationships				
Unit-III	A) Interpersonal Skills and Leadership (10 HRS) Communication Skills and Conflict Resolution Giving and Receiving Constructive Feedback Influence and Persuasion in Team Settings Leading with Emotional Intelligence Case Studies: EI in Engineering Teams and Leadership Scenarios B) Write Emotional Intelligence & present seminar.				
References	Sr. No.	Title	Author	Publication	Edition
	1	Emotional Intelligence: Why It Can Matter More Than IQ	Daniel Goleman	Bantam Books	10th Edition
	2	Working with Emotional Intelligence	Daniel Goleman	Bantam Books	1st Edition
	3	Emotional Intelligence 2.0	Travis Bradberry, Jean Greaves	TalentSmart	Revised Edition

FACULTY OF ENGINEERING AND TECHNOLOGY Syllabus Structure of SY Electrical Engineering / Electrical and Electronics Engineering/ Electrical, Electronics and Power (EE/EEE/EEP) SEM-IV	
Course Code: PCCT-1 Course: AC Machines Credits: 3 Theory: 3Hrs./ Week;45 Hrs / Semester	
Prerequisites	Knowledge of Basic Electrical Engineering And DC Machine & Transformer
Objectives	1. Gain an understanding of the fundamental principles of alternating current (AC) and how it applies to electrical machines. 2. Learn about the working principles of AC machines (motors and generators), including concepts like magnetic fields, electromagnetic induction, and torque production. 3. Understand the construction of these machines, including components like the stator, rotor, armature, and field windings. 4. Study the different types of AC machines, such as synchronous machines and induction machines (both motors and generators). 5. Distinguish between single-phase and three-phase AC machines.
Course Outcomes	1. Understand the concept of AC machines and their industrial applications 2. Apply the concept of equivalent circuit of machines in different application 3. Analyze the numerical methods in ac machines 4. Evaluate the performance analyses of different AC machines.
Unit-I	A : Three phase Induction Motor Three phase Induction Motor Construction & types of 3 ph. Induction motors, torque equation, starting torque, running torque, condition of maximum torque ,torque slip characteristics, Need of starters for 3 phase Induction motors, types of starters, Speed control methods from stator side (Stator voltage control ,Stator Frequency control, Pole changing) & rotor side (rotor resistance control), Applications of 3 ph. Induction motors. (10Hrs)
	B : Equivalent circuit analysis of three phase induction motor Equivalent circuit analysis of three phase induction motor Losses & efficiency of 3 phase induction motor, power flow diagram with numerical treatment, No load & blocked rotor test, equivalent circuit of 3 phase induction motor, Phasor diagram of 3 phase induction motor, performance of 3 phase induction motor using circle diagram, Cogging & crawling of 3 phase induction motor (6 Hrs)
	A : Single Phase Induction Motor Single Phase Induction Motor Construction, Working and types of single phase induction motors (Split phase, capacitor start/run, shaded pole motors), Double field revolving theory, Characteristics & Applications (8 Hrs)

Unit-II	B : Alternator Alternator Construction, principle of operation of three phase alternator, emf equation, parameters of armature winding, armature reaction, concept of synchronous reactance and synchronous impedance. Equivalent circuit of 3 phase alternator, alternator on load (resistive, inductive & capacitive) (8 Hrs)
---------	---

Unit-III	A : Equivalent Circuit Analysis Of Alternator Equivalent circuit analysis of Alternator OC test & SC test on 3 Phase alternator, short circuit ratio, voltage regulation methods (emf, mmf, zero power factor and direct loading method) with numerical treatment, Losses and efficiency, power flow diagram, need of parallel operation, conditions for parallel operation, synchronizing procedures, hunting and oscillations in alternators (7 Hrs)
	B : Synchronous motor Synchronous motor Synchronous motor, starting methods, Phasor Diagram, Effect of excitation on power factor and armature current, V and inverted V Curves, Operation of Synchronous motor as Synchronous Condenser, Applications of three phase synchronous motor. (6 Hrs)

References	Sr. No.	Title	Author	Publication	Edition
	1	Electrical Machines	S. K. Bhattacharya	Tata Mc-Graw-Hill publication	3 th Edition
	2	Electrical Machines	I. J. Nagrath, D. P. Kothari	Tata McGraw Hill publication	4 th Edition
	3	Electric Machinery	A. E. Fitzgerald	S. Chand Publishing	4 th Edition
	4	Theory of AC machines	A. S. Langsdorf	Mc-Graw Hill publications	5 th Edition
	5	Design of Brushless Permanent Magnet motors,	J. R. Hendershot and T. J. E. Miller	Magna Physics Publishing and Clarendon press.	1994 Edition

FACULTY OF ENGINEERING AND TECHNOLOGY Syllabus Structure of SY Electrical Engineering / Electrical and Electronics Engineering/ Electrical, Electronics and Power (EE/EEE/EEP) SEM-IV	
Course Code: PCCP-1 Course: AC Machines Lab Credits: 1 Theory: 2Hrs./ Week; 30 Hrs / Semester	
Prerequisites	Machines Fundamental
Objectives	1. Students will be able to explain the basic principles and concepts of AC machines, including electromagnetic induction, alternating current, and the relationship between voltage, current, and magnetic fields in machines. 2. Learn about the working principles of AC machines (motors and Students will identify and distinguish between various types of AC machines, such as single-phase and three-phase machines, synchronous and asynchronous machines, and their specific applications.
Course Outcomes	1. Students will be able to make appropriate connections for testing of AC machines. 2. Students will be able to deduce conclusions about the performance using obtained readings.
Experiments	Any 12 practical to be conducted on 1. Identification of different parts of a three phase squirrel cage and slip ring induction motor, interpretation of the nameplate of three phase induction motor and reversal of the direction of rotation 2. Starting of a three-phase induction motor using (a) auto transformer (b) DOL starter (c) star-delta starter 3. Determination of efficiency & speed regulation of 3 phase induction motor by direct loading method 4. Determination of circle diagram parameters of 3 Phase induction motor by conducting No Load & Blocked Rotor Tests. 5. Speed control of a three-phase slip ring induction motor by varying rotor resistance. 6. Study of starters for 3 Phase induction motors. 7. Speed control methods of 3 Ph.IM. (Stator Side). 8. Speed control methods of 3 Ph.IM. (Rotor Side). 9. Determination of efficiency & speed regulation of 1-phIM 10. Determination of Voltage regulation of an alternator by EMF method. 11. Determination of Voltage regulation of an alternator by MMF method 12. Determination of Voltage regulation of an alternator by ZPF method. 13. Determination of X_d and X_{q0} of an Alternator by Slip test

	14. Determination of efficiency of synchronous motor by direct loading method 15. Determination of efficiency and regulation of Alternator by direct loading method
--	--

Faculty of Science & Technology Syllabus of S. Y. B.E./B.TECH.(Computer Science and Engineering; AI&ML; Electrical Engineering, Artificial Intelligence, Information Technology, Electrical and Electronics Engineering, Electronics and Telecommunication Engineering) (Semester III)	
Course Code: PCCT-2 Course: Electrical Power Transmission and Distribution Credits: 2 Theory :2Hrs./ Week ; 30 Hrs / Semester	
Prerequisites	Basic Sciences
Objectives	0. To provide knowledge of some electrical power transmission and distribution. 1. To understand configuration of transmission network and know its types. 2. To study Insulating materials and their applications. 3. To expose the students to working of transformer and their applications. 4. To introduce basic aspects of transmission and distribution system.
Course Outcomes	Understand Power System Components. Analyze Transmission Line Parameters. Design basic digital circuits. Evaluate Power Flow and Voltage Regulation. Apply Fault Analysis and Protection Techniques. Design and Plan Distribution Systems.
Unit-I	A) BASIC CONCEPTS (06 HRS.) Different types of distribution systems ring main , radial distribution, types & choice of distribution system conductor, types of loads in transmission and distribution systems, Load curve , load duration curve, demand factor, load factor, diversity factor, load forecasting concept, Tariff: Residential, commercial, H.T., L.T. Time of Day tariff, Incentives and penalties.
	B) ELECTRICAL EQUIPMENT'S IN SUB-STATIONS (04 HRS.) Descriptive treatment of ratings, Special features, field of use of equipment's like transformers, bus-bars, voltage regulators, switches and isolators, reactors, Control panels, metering, power supplies like station transformers, storage batteries and other control room equipment's in sub-stations
Unit-II	A) OVERHEAD LINE INSULATORS (04 HRS.) Types of insulators, pin type, suspension type, shackle type, strain type insulators, voltage distribution along string of suspension insulators, distance of insulators string efficiency, Equalization of potential across each unit.
	B) DESIGN OF OVERHEAD LINES: (06 HRS.) Line supports, spacing between the conductors, length of span, calculation of sag, equal and unequal supports, effect of ice and wind loadings. UNDERGROUND CABLE: Classification, Construction of cable, XLPE cables, insulation resistance, capacitance, dielectric stress in single core/multi core cables, cable faults and location of faults.

Unit-III	A) CONSTANTS OF TRANSMISSION LINE (10 HRS) Inductance Calculation Skin effect and its properties, proximity effect, Ferranti effect, Corona effect, concept of G.M.R. and G.M.D, inductance of single phase two wire line, inductance of parallel circuit three phase line, three phase line with equilateral spacing, unsymmetrical spacing. Capacitance Calculation capacitance of three phase line with equilateral spacing, capacitance of parallel circuit three phase line with equilateral spacing, unsymmetrical spacing, capacitance of single phase line with earth effect and without effect of earth's surface on electric field, calculation of capacitance to be done with and without transposition.				
References	Sr. No.	Title	Author	Publication	Edition
	1	Power System Engineering	D.P. Kothari and I.J. Nagrath	McGraw Hill Education	3rd Edition
	2	Electrical Power Systems	C.L. Wadhwa	New Age International Publishers	6th Edition
	3	Modern Power System Analysis	D.P. Singh and R.P. Gupta	S. Chand Publishing	2 nd Edition
	4	Electrical Power Distribution Systems	V.K. Mehta and Rohit Mehta	S. Chand Publishing	Revised Edition (2020)
	5	A Course in Electrical Power	J.B. Gupta	S.K. Kataria & Sons	4 th Edition.

Faculty of Engineering & Technology Syllabus of S.Y.B.E (Computer Science and Engineering; AI & ML; Electrical and Electronics Engineering) (Semester IV)		
CourseCode:ESC130 Course: Electrical Power Transmission and Distribution LAB Credits: 1 Practical:02Hrs./Week; 30Hrs/Semester		
Prerequisites		Proficiency in Using Laboratory Tools, Basic Technical Documentation Reading Skills, Basic Troubleshooting Skills
Objectives	1. 2. 3. 4.	Gain practical experience working with electrical components and circuits. Learn to identify, handle, and understand the behavior of distribution and transmission components. Develop skills in building and assembling electrical circuits & Networks. Learn to use measuring instruments like multimeter, voltmeter, Ammeter.
Course Outcomes	1. 2.	Conduct experiments on Transmission & Distribution components and verify their characteristics. Demonstrate experiments on Transmission & Distribution circuits.
List of Experiments	1. 2. 3. 4. 5. 6. 7. 8. 9. 10. 11.	PART: A Study of Transmission Sub-station & Drawing sheet of 132kV or 400kV sub-station. Study and Drawing sheet of pin type, strain type, shackle type insulators. Measurement of ABCD parameters of a transmission line (on model kit). Measurement of capacitance of std. cable sample of 1-mtr length, using Shearing Bridge. Meggering of armored cable with 2500V megger. Identification of faults in cable. Mat-lab simulation of transmission line model. Mat-lab simulation of faults in transmission line model. Study of ACSR conductors. Evaluation of line parameters of 'π' model. (Expt. On model kit.) Evaluation of line parameters of 'T' model. (Expt. on model kit.) PART: B Visit to transmission station and submit the report individually

FACULTY OF ENGINEERING AND TECHNOLOGY		
Syllabus Structure of SY Electrical Engineering / Electrical and Electronics Engineering/ Electrical, Electronics and Power (EE/EEE/EEP) w.e.f.2025-26		
(As Per NEP-2020)		
Course Code: PCCT-3 Course: Network Analysis		
Semester-IV		
Credits: 2 (Theory); 1(Practical)		
Teaching Scheme	Examination Scheme	
Lecture: 2 hrs./week Practical: 2 hr./week	Mid Semester Exam: 20 Marks End Semester Exam: 30 Marks	
Prerequisites		
Knowledge of - Basics of Electrical Engineering. and Engineering Mathematics-I		
Objectives		
1. Provide the student with a comprehensive understanding of the basic law of electric circuit & theories.		
2. To make the students capable of analyzing any given electrical network.		
3. To learn about the use of mathematics, Laplace Transform & differential equations for network analysis.		
4. To make the students learn how to represent an electrical network in terms of different parameters.		
Course Outcomes		
CO1 Solve Circuits using Topology and circuit reduction techniques.		
CO2 Analyze the AC circuit using different network theorems.		
CO3 Analyze the circuit for steady state and transient response in time domain		
CO4 Analyze and evaluate transient response, Steady state response, network functions in S-domain		
CO5 Express given Electrical Circuit in terms of A,B,C,D and Z,Y Parameter model and solve the circuits.		
ModuleNo.	Topics / actual contents of the syllabus	Contact Hours
Unit-I	Basic Concepts: Electrical parameters, Voltage and current sources(Dependent and Independent), Classification of electrical elements, Dot convention for coupled circuits, Concept of duality and dual networks. Network Topology Network Theorems: (Application to networks with all types of sources.) Node, Mesh, Super mesh & Super node analysis, Superposition, Thevenin's and Norton, Reciprocity, Substitution theorems, Millman's Theorem, Maximum power theorem, Compensation Theorem	(10 hrs)
Unit-II	Solution of Network Equations: Steady state and transient solution, Forced and free response, Time constants, Physical and mathematical analysis of circuit transients, Laplace Transform Method: Solution of differential equations and network equations using Laplace transform method inverse Laplace transform, Transformed networks with initial conditions analysis of electrical circuits with applications of step, impulse and ramp functions, shifted and singular functions, Theconvolution integral Laplace transform of various periodic and non-periodic waveforms.	(10 hrs)
Unit-III	Two Port Networks: Z,Y and transmission parameters H parameters, Interrelations between these parameters,Transfer function, Concepts of poles and zeros, Transform impedance, Transform admittance, Concept of complex frequency, Driving point and transfer impedance and admittances.	(10 hrs)

Reference Books and Text Books:	
25.	Smarajit Ghosh, Network Theory: Analysis And Synthesis 1st Edition, Phi Learning Pvt. Ltd
26.	C. L. Wadhwa, Electrical Circuit Analysis: Including Passive Network Synthesis, 2nd edition, New AGE INTERNATIONAL PUBLISHERS LTD.-NEW DELHI
27.	M. Musa, Matthew N. O. Sadiku, Charles K. Alexander, Applied Circuit Analysis, 2nd edition, Mcgraw Hill Education
28.	William H. Hayt Jr., Jack E. Kemmerly, Steven M. Durbin, Engineering Circuit Analysis, 6th TMH.
29.	M.E. Van Valkenburg, Network Analysis, 2nd edition, Prentice Hall.
30.	Boylestad Robert L. Charles E., Introduction to Circuit Analysis, Merrill Publishing Company.
31.	John R. O Malley, Circuit Analysis, Prentice Hall.

Lab Network Analysis

Teaching Scheme Lectures : 2 Hrs/Week

Laboratory course outcomes

After completion of the course students will able to:

32. CO1 Apply various basic laws and theorems of electrical circuit
33. CO2 Understand effects of Initial and final conditions on networks.
34. CO3 Understand and examine behavior of the network for the applications of Step, Impulse and Ramp functions.
35. CO4 Explain the fundamental principle Fourier transform of Waveform Analysis.
36. CO5 Analysis of electrical networks using two port networks concept and Concepts of Poles and Zeros

List of Experiments (any eight or more by simulation or Hardware)

Sr No.	Name Of Experiment
1	Network Theory
2	Verification of Superposition Theorem
3	Verification of Thevenin's Theorem
4	Verification of Maximum Power Transfer Theorem
5	Verification of Reciprocity Theorem
6	Study of Step Response of R-L Network
7	Study of Step Response of R-C Network
8	Study of Time Response of R-L-C Network
9	Waveform Analysis by Fourier Methods

FACULTY OF ENGINEERING AND TECHNOLOGY Syllabus Structure of SY Electrical Engineering / Electrical and Electronics Engineering/ Electrical, Electronics and Power (EE/EEE/EEP)	
Course Code: MDMT-1 Course: EV Technology and Components Credits: 2 Theory :2Hrs./ Week ; 30 Hrs / Semester	
Prerequisites	Basics of Electrical and Electronics, Engineering Chemistry, Engineering Systems.
Objectives	To understand the basic technologies used in e-vehicles and the necessary advancements in it
Course Outcomes	1. Understand the basics related to e-vehicle. 2. Identify and describe key EV components including electric motors, batteries, power electronics, and controllers. 3. Analyze the performance characteristics and efficiency of EV drivetrain systems. 4. Understand the architecture and working principles of electric vehicles.
Unit-I	Electric Propulsion unit: Introduction to electric components used in hybrid and electric vehicles, Configuration and control of DC Motor drives, Configuration and control of Induction Motor drives, configuration and control of Permanent Magnet Motor drives, Configuration and control of Switch Reluctance Motor drives, drive system efficiency.
Unit-II	Energy Storage: Introduction to Energy Storage Requirements in Hybrid and Electric Vehicles, Battery based energy storage and its analysis, Fuel Cell based energy storage and its analysis, Super Capacitor based energy storage and its analysis, Flywheel based energy storage and its analysis, Hybridization of different energy storage devices. Sizing the drive system: Matching the electric machine and the internal combustion engine (ICE), Sizing the propulsion motor, sizing the power electronics, selecting the energy storage technology, Communications, supporting subsystems.
Unit-III	Energy Management Strategies: Introduction to energy management strategies used in hybrid and electric vehicles, classification of different energy management strategies, comparison of different energy management strategies, implementation issues of energy management strategies. Case Studies: Design of a Hybrid Electric Vehicle (HEV), Design of a Battery Electric Vehicle (BEV).
10	Verification of Two Port Network

Sr. No.	Title	Author	Publication	Edition
1	Design Fundamentals	Iqbal Hussein, Electric and Hybrid Vehicles	CRC Press, 2003	2003
2	Electric and Fuel Cell Vehicles: Fundamentals,	Mehrdad Ehsani, Yimi Gao, Sebastian E. Gay	-	2004
3	Electric Vehicle Technology	James Larminie, John Lowry	S. Chand Publishing	2003